Algorithm 1 Overall Risk-aware Decision Procedure

Require: Scenario x_t, L0 store $\mathcal{M}^{L0}$, L2 store $\mathcal{M}^{L2}$, limits τ_R, τ_Q
Ensure: Approved action a_t^{app} or $\perp$, and decision trace $\mathcal{T}_t$

for $k \in \{\mathrm{adv}, \mathrm{cri}, \mathrm{dec}\}$ **do**
    $\mathcal{B}_t^{(k)} \leftarrow \textsc{RetrieveRiskMemory}(x_t, \mathcal{M}^{L0}, k, K, B_k)$
end for
$\mathcal{Q}_t \leftarrow \textsc{MatchRiskRules}(x_t, \mathcal{M}^{L2})$
$\rho_t^{rule} \leftarrow \textsc{ResolveRuleConflicts}(\mathcal{Q}_t, \mathcal{A}_t)$
if $\rho_t^{rule}.\mathrm{status} = \mathrm{conflict}$ **then**
    $d_t^{conf} \leftarrow \textsc{BuildConflictDecision}(\rho_t^{rule})$
    $a_t^{\mathrm{app}} \leftarrow \textsc{SafetyGuard}(x_t, d_t^{conf}.\mathrm{action})$
    $\mathcal{T}_t \leftarrow \textsc{BuildConflictTrace}(\mathcal{B}_t, \mathcal{Q}_t, \rho_t^{rule}, d_t^{conf}, a_t^{\mathrm{app}})$
    return $a_t^{\mathrm{app}}, \mathcal{T}_t$
end if
$d_t^{\mathrm{adv}} \leftarrow \textsc{TaskAdvocate}(x_t, \mathcal{B}_t^{(\mathrm{adv})}, \mathcal{Q}_t)$
if $\neg\textsc{ValidReport}(d_t^{\mathrm{adv}})$ **then**
    return $\perp, \mathrm{error}_{adv}$
end if
$d_t^{\mathrm{cri}} \leftarrow \textsc{RiskCritic}(x_t, \mathcal{B}_t^{(\mathrm{cri})}, \mathcal{Q}_t)$
if $\neg\textsc{ValidReport}(d_t^{\mathrm{cri}})$ **then**
    return $\perp, \mathrm{error}_{cri}$
end if
$d_t^{\mathrm{dec}} \leftarrow \textsc{SafetyDecision}(x_t, \mathcal{B}_t^{(\mathrm{dec})}, \mathcal{Q}_t, d_t^{\mathrm{adv}}, d_t^{\mathrm{cri}})$
if $\neg\textsc{ValidReport}(d_t^{\mathrm{dec}})$ **then**
    return $\perp, \mathrm{error}_{dec}$
end if
$a_t^{\mathrm{LLM}} \leftarrow d_t^{\mathrm{dec}}.\mathrm{action}$
$\mathcal{A}_t^{\mathrm{safe}} \leftarrow \textsc{FilterUnsafeActions}(x_t, \mathcal{Q}_t)$
for all $a \in \mathcal{A}_t^{\mathrm{safe}}$ **do**
    $u_a \leftarrow \textsc{TaskUtility}(a)$
    $r_a \leftarrow \textsc{ActionRisk}(x_t, a, \mathcal{B}_t^{(\mathrm{dec})})$
    $q_a \leftarrow \textsc{ActionUncertainty}(x_t, a, \mathcal{B}_t^{(\mathrm{dec})}, d_t^{\mathrm{adv}}, d_t^{\mathrm{cri}}, d_t^{\mathrm{dec}})$
end for
$\mathcal{E}_t \leftarrow \{a \in \mathcal{A}_t^{\mathrm{safe}} \mid r_a \le \tau_R \wedge q_a \le \tau_Q\}$
if $\mathcal{E}_t \neq \emptyset$ **then**
    $a_t^* \leftarrow \textsc{DeterministicBest}(\mathcal{E}_t, u, r, q)$
else
    $a_t^* \leftarrow \textsc{ConservativeFallback}(\mathcal{A}_t)$
end if
$a_t^{\mathrm{app}} \leftarrow \textsc{SafetyGuard}(x_t, a_t^*)$
$\mathcal{T}_t \leftarrow \textsc{BuildDecisionTrace}(\mathcal{B}_t, \mathcal{Q}_t, \rho_t^{rule}, d_t^{\mathrm{adv}}, d_t^{\mathrm{cri}}, d_t^{\mathrm{dec}}, a_t^{\mathrm{LLM}}, a_t^*, a_t^{\mathrm{app}})$
return $a_t^{\mathrm{app}}, \mathcal{T}_t$